Troubleshooting the Organic Garden: Common Issues and Simple Solutions

Exhausted Soil, Insects, Critters, Disease, Inappropriate Watering, Nutrient Deficiency, Blossom Drop

You can grow any food plants that will fit the growing time, seasonal temperatures, soil fertility, and water & gardener availability. Organic "Garden Soil" or "Potting Mix" free from contaminants contributes to the health of both the plants and the gardener.

Every growing season is a balancing act for nature. Nature out-of-balance encourages invasions of insects, critters, and plants.

Do not expect gardening perfection as there are too many variables which change from year to year: weather (temperature, sun, wind, rain, severe storms), soil quality, compost quality, mulch affect, insects, viruses, fungi, bacteria, weeds, critters, thieves, vandals, and gardener's practices. What worked well last season may not be so terrific this season. Celebrate the successes and next season adjust for the failures. We have had to do this every year for 50+ years.

Maintaining an Urban Organic Food Garden

- 1st – maintain <u>healthy soil</u> & healthy plants.
- 2nd -- encourage helpful insects & critters.
- 3rd – <u>manage</u> the harmful insects, critters, diseases, invasive plants, climate, & people.

Basic Facts

- Gardens have helpful & harmful
  - Fungi
  - Bacteria
  - Insects
  - Plants
  - Animals
  - Weather
  - People

Bacteria

Weeds

Stink Bugs

Racoon

Voles

- Plants do well in some soils but poorly in others.
- Soil composition affects taste.
- **Note:** weakened plants attract feeding insects as per nature's design! **Healthy plants fight back & can tolerate a few bugs**.

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Exhausted Soil

- Exhausted soil shows up as lackluster performance in plants, weakened plants, and the onset of more than the usual diseases and troublesome insects.
- Best possible soil amendment for food plants and flowers is **compost**! It nourishes the soil (slow-release of plant nutrients from A-Z), fights disease over time, balances pH, retains moisture, and conditions soil structure to be more friable (more workable).
- Alternatives are cut-and-drop the soft parts of spent plants (veg, flowers, & seedless weeds), purchased compost, or well composted farm animal manure (chicken, sheep, cow, horse).
- Then, add a natural mulch (shredded leaves, grass clippings, stones, …).

Clay (thin line but nutrient rich)

Sand (nutrient poor; drains too fast)

Compost (rich organic matter)

Insects

Mud Wasp Nest

Helpful

Pollinators: bees, moths, flies, mosquitos, ….

Insect Predators: Assassin bug, Big Eyed bug, Damsel bug, Damselflies, Dragonflies, Fireflies, Ground beetles, Hover Flies (Syrphid), Lacewing flies, Ladybugs, Mantids, Minute Pirate bug, Robber Flies, Rove beetles, Soldier beetles, Spiders, Predatory Stink bug, Tachinid Flies, Tiger beetles, Wasps.

Lady Bug Larva takes Fly

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Grapevine Beetle

Harmful

Aphids
Beetles (Flea, Colorado, Cucumber, Japanese)
Caterpillars
Cutworms
Earwigs
Leaf Miners
Slugs & Snails
Vine borers

Critters

Harmful

- Deer (eat leafy greens [my squash, cucumber, & beans])
- Groundhogs (eat leafy greens)
- Mice (eat seeds)
- Rats (omnivores)
- Rabbits (eat leafy greens)
- Racoons (eat corn, compost items, and green bin contents)
- Squirrels (eat & plant seeds & dig holes)
- Voles (eat grass, roots, bulbs, seeds, ….)

Helpful

- Bats (eat insects)
- Birds (all young birds get protein)
- Frogs (pond & tree versions eat insects)
- Racoons (eat lawn grubs)
- Shrews (eat insects, beetles, worms, etc.)
- Skunks (eat lawn grubs)
- Snakes (eat rodents)
- Toads (eat insects)

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Garden Helpers

Hornet takes Caterpiller

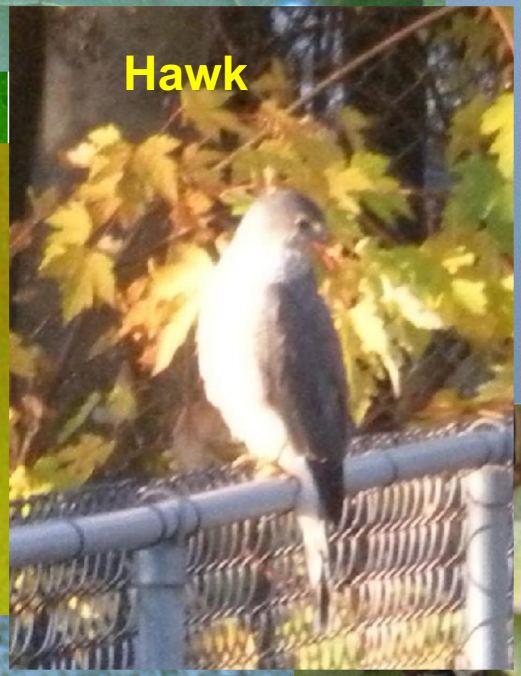
Spider

Parasitic Wasp

Lacewing

Bat

Rove Beetle


Hawk

Skunk & skunk foraging

Firefly

Cdn Tree Frog

Diseases: Bacteria, Fungi, & Viruses

Bacteria start as skin/rind dimples and create serious discolouration on leaves and fruit.
Fungi have furry or fruiting bodies & often create dark or white colouration on leaves.
Viruses produce unusual leaf colour (e.g., motley) & odd structural variations in leaves.

Bacteria on produce (e.g. tomato tops, fruit that touches ground soil or has soil splash), appears as strong colour variation & wasted areas. Water helps it to stick. Mulch to protect fruit and leaves from soil; water plant **roots only-** not the leaves or the fruit. A thirsty plant needs a drink, not a bath. Compost compromised material.

Tomato blight (both early & late) on leaves & stalks: remove & green bin **all** affected parts + fallen debris; rotate plants to new locations **and new soil** for the next 2 to 3 seasons. Apply significant amounts of compost to the former area.

Powdery Mildew (white or grey) on leaves – spray milk & water 50/50, or baking soda & water - do not rinse. Slows, does not stop. Safe to compost at end of season.

Mosaic virus via sap sucking insects (e.g., aphids & cucumber beetles) - remove entire plant and yard waste, solarize, or torch it. Do not compost any parts.

Squash Bacteria; Tomato Blight; Powdery Mildew; Corn Smuts; Mosaic Virus

Under, Over, & Inappropriate Watering

Under watering causes plants to wilt or droop as water in the plant makes leaves and stalks firm – like air in a balloon. **Water the roots only**, and the plant should recover in an hour or two.

Over watering or lack of good drainage leads to drowning – plants need moisture **and air** at the roots and too much water eliminates air. Leaves will turn yellow, usually from the bottom first, but not always. This problem is slow to appear and plants are slow to recover since the damage is more severe. Excess water has to be removed as soon as possible. Re-potting may be necessary.

Inappropriate watering - "Blossom end rot" in tomatoes and peppers is due to inconsistent watering which stops the uptake of calcium to the fruit. These plants do not swim so soil must be consistently moist but well drained (no water drips from hand squeezed soil); however, they must not be allowed to dry out. This problem is easy to identify and treat. **Many Herbs** need wet and dry periods to intensify the oils we enjoy. They must dry out. **Constant moisture** is needed for tomatoes, salad greens, cilantro, basil, while most other veg can tolerate **mild wet and dry** periods. The deeper the root the better chances of finding moisture.

Nutrient Deficiencies

Nutrient deficiency shows up as an odd colouration (usually yellow, tawny, or purplish) on either the edges or the centers of leaves around the veins, and plant growth may be stunted. Trying to figure out which nutrient is missing is a waste of your time – many deficiencies show up as yellow, including drowning and spent plants.

Solution: add 4 cm+ of compost or composted farm animal manure (not pig), or add organic liquid or granulated fertilizer according to instructions. Recovery will be slow until water soluble nutrients replenish the soil and reach the plant.

Note: – Excess nutrients burns plants. Most excesses get rained/watered away and wasted. Too much nitrogen (N) grows large green plants with little to no fruit at all (e.g., too much Mushroom Compost or fertilizer with high N). High NPK numbers are good for a lawn or meadow – **not** your food and flower gardens (e.g. use 20-20-20 for millions of plants like your lawn; use decimal points for tens of plants!)

Plant Health Check

- Look over and under leaves for insects, insect eggs, webbing, disease, unusual structures, and/or colour (usually motley &/or not normal).
- Remove wilted material. If diseased, place it in yard waste; if not, compost it. Wilted material can infect other parts of a plant or neighbouring plants (e.g., tomato blight) or hasten the decomposition of other leaves (e.g., Lettuce).
- Remove competing plants if the plants are interfering with the health of the edibles (e.g., weeds, tree/shrub roots, veg & flower bullies).
- Trim plants to improve ventilation: reduce humidity, & avoid the growth of bacteria & fungi.
- Add extra supports to keep plants from touching the soil and trim off any lower branches or leaves that touch the soil. Lower leaves can pick up soil born diseases, and they provide a stairway for slugs and snails. Add Mulch.
- **Option, but recommended**: after a healthy amount of growth, trim off new flowers to reduce the energy drain on the plants (e.g., stop continued tomato or squash production to concentrate energy, growth and flavour in the available fruit; remove potato flowers, herb flowers, …).

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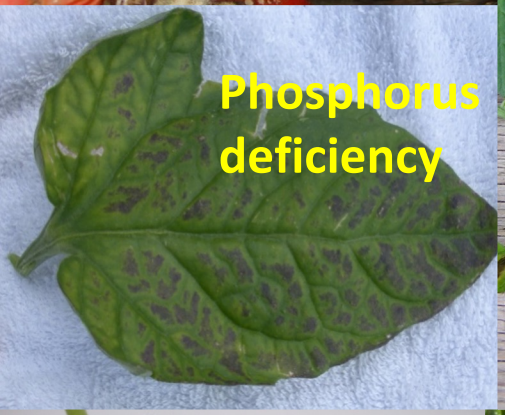
Plant Health



Garden Maintenance

Attract & Keep Pollinators & Insect Predators; **Deter** critters with barriers – nets, row covers, or fencing. **Aerate/loosen** container soil approx. once/mo. Caution: do not sever roots (recommend Jekyll weed fork). **Thin dense foliage** for better air circulation. **Water** with watering can so you can check <u>each</u> plant for issues and save water. Weed Weekly.

Inter-planting for pollinators and isolation for rotation, invasive plants (e.g., mint, oregano, …), & plant bullies (e.g., tomato).

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Plant Rotation offsets nutrient loss, disease, insect eggs, & insect larvae.

Heavy feeders

The larger the plant & fruit, the heavier the feeder and the greater depletion of soil nutrients. (e.g. Corn, Cabbage Family, Rhubarb, All Vine plants)

Where possible it is best to move plants (not perennials) to new areas/rows seasonally to reduce disease, insects, & nutrient exhaustion.

Solutions for box beds, & elevated beds: rotate troubled plants into satellite containers with new soil for 2 seasons.

-- Try different plants in the beds for 2 seasons and forget the troubled ones.

-- Regardless, Add compost, compost, & more compost – no such thing as too much compost!

Rotation for Plots and Raised beds

Effects of No Rotation/ No Diversity:

- Build up of insects and diseases in the soil.
- Soil nutrient exhaustion.
- Slow nutrient recovery even with compost.

Benefits of Rotation & Diversity:

- Plants take different nutrients at different times and at different depths, so change plants!
- Insects and diseases have to look elsewhere for their favourite plants.
- Compost has more time to work on diseases and pH.

Suggestions:

Divide the available growing spaces into distinct areas and rotate plantings each season into a new area. Be sure to add compost/composted farm animal manure twice a season – do not rake under (fall and mid-season are recommended times).

At the very least, consider relocating plants which seem troubled each season and more susceptible to insects and disease.

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Greens use N
(moderate feeders)

(Leaf Lettuce)

Fruits use PK: Corn, Vines
(heavy feeders)

(Celebrity &
Lemon Boy)

(Sugar Snaps)

Legumes put N into the soil:
Peas, beans, soybeans….

(Danvers)

Roots use P (light feeders)

Note: this grouping of veg might help with your rotation decisions.

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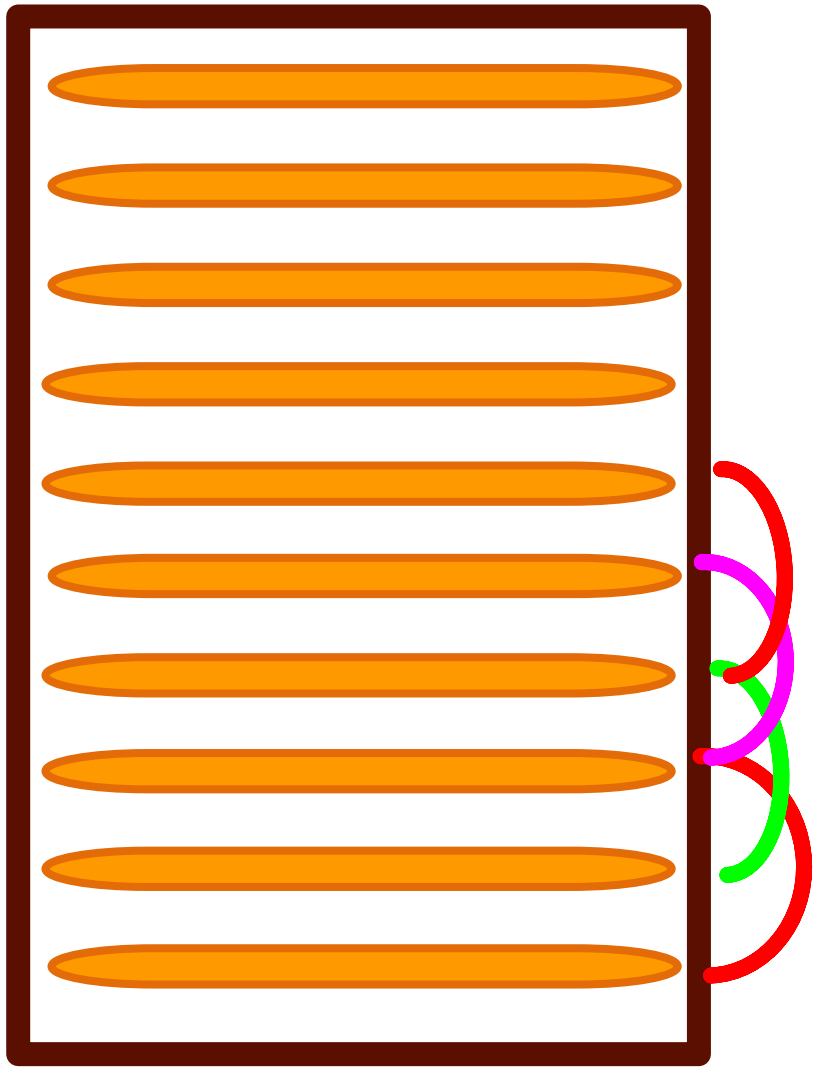
Narrow plot or raised bed

Tank Tread
Move plants 1, 2, or 3 rows from the last season's position.

Farther from their last position is better than nearer.



Note:
rotation is easier with permanent rows but some general markers and phone photos can help with next season's placements in plots and raised beds.

Pro-active Plant Stress & Disease Management

to reduce bug attacks & blossom drop.

1. **Reduce weed pressure** (remove weeds & add natural mulch)
2. **Follow watering guides** (water roots only x2 -- no swamps, ever!)
3. **Rotate plants to new areas each season** – or at least relocate the troubled ones.
4. **Ensure sun requirements** (trim bushy plants, & sun boost or sun screen as needed – **NB** leaves protect fruit from sun scald!)
5. **Reduce wind destruction, wind drying & air borne diseases** (set a windbreak – e.g., tall/bushy plants, poly tunnel, row cover)
6. **Keep leaves & fruit from soil & soil splash** (trim lower leaves and add a soil cover: mulch of leaves, cardboard, or purchased soil cover)
7. **Aerate all dense soils for air & water penetration** (don't cut roots)
8. **Maintain air circulation to reduce humidity** (thin bushy plants)
9. **Fertilize twice a year (**fall or early spring & mid-season) with compost, composted farm animal manure, liquid or granulated fertilizer.

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Reactive Plant Stress & Disease Management

Check individual plants often to discover any issues before they get out of control (e.g., water the base of plants with a watering can to get up close and personal to inspect each plant).

#1 Remove diseased parts ASAP [leaves & fruit]

#2. Remove all disease debris from soil [do not compost]

#3. Clean garden tools to avoid transfer of disease [hot soapy water, rubbing alcohol, hand sanitizer, or diluted bleach]**.**

#4. Spray powdery mildew with water & baking soda or water & milk 50/50 [slows, doesn't cure]

#5. Spray aphids, insect eggs, and insect larvae with soapy water and rinse after 10 minutes – but, drown adult beetles in a dish of soapy water.

#6. Add compost NB: we "mulch" with compost, but compost is not a mulch. Compost is a highly nutritious buffet for all plants, including weeds! Cover compost additions with organic mulch.

Reminder: Compost neutralizes disease over time, balances pH, adds nutrients A to Z, retains moisture, improves soil structure, and makes soil friable (loose) – best fertilizer your garden could have!

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A Happy Garden Makes a Gardener Happy.
Keep calm and carry on urban organic food & flower gardening.
The many benefits are worth it, despite any setbacks or sacrifices to nature!